

Chapter 2

Ergodicity, Recurrence and Mixing

In this chapter the basic objects studied in ergodic theory, measure-preserving transformations, are introduced. Some examples are given, and the relationship between various mixing properties is described. Background on measure theory appears in Appendix A.

2.1 Measure-Preserving Transformations

Definition 2.1. Let $(X, \mathcal{B}, \mu)$ and $(Y, \mathcal{C}, \nu)$ be probability spaces. A map ϕ from X to Y is *measurable* if $\phi^{-1}(A) \in \mathcal{B}$ for any $A \in \mathcal{C}$, and is *measure-preserving* if it is measurable and $\mu(\phi^{-1}B) = \nu(B)$ for all $B \in \mathcal{C}$. If in addition ϕ^{-1} exists almost everywhere and is measurable, then ϕ is called an *invertible measure-preserving map*. If $T : (X, \mathcal{B}, \mu) \rightarrow (X, \mathcal{B}, \mu)$ is measure-preserving, then the measure μ is said to be *T-invariant*, $(X, \mathcal{B}, \mu, T)$ is called a *measure-preserving system* and T a *measure-preserving transformation*.

Notice that we work with pre-images of sets rather than images to define measure-preserving maps (just as pre-images of sets are used to define measurability of real-valued functions on a measure space). As pointed out in Example 2.4 and Exercise 2.1.3, it is essential to do this. In order to show that a measurable map is measure-preserving, it is sufficient to check this property on a family of sets whose disjoint unions approximate all measurable sets (see Appendix A for the details).

Most of the examples we will encounter are algebraic or are motivated by algebraic or number-theoretic questions. This is not representative of ergodic theory as a whole, where there are many more types of examples (two non-algebraic classes of examples are discussed on the website [81]).

* In this measurable setting, a map is allowed to be undefined on a set of zero measure. Definition 2.7 will give one way to view this: a measurable map undefined on a set of zero measure can be viewed as an everywhere-defined map on an isomorphic measure space.

We define the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ to be the set of cosets of $\mathbb{Z}$ in $\mathbb{R}$ with the quotient topology induced by the usual topology on $\mathbb{R}$. This topology is also given by the metric

$$d(r + \mathbb{Z}, s + \mathbb{Z}) = \min_{m \in \mathbb{Z}} |r - s + m|,$$

and this makes $\mathbb{T}$ into a compact abelian group (see Appendix C). The interval $[0, 1) \subseteq \mathbb{R}$ is a fundamental domain for $\mathbb{Z}$: that is, every element of $\mathbb{T}$ may be written in the form $t + \mathbb{Z}$ for a unique $t \in [0, 1)$. We will frequently use $[0, 1)$ to define points (and subsets) in $\mathbb{T}$, by identifying $t \in [0, 1)$ with the unique coset $t + \mathbb{Z} \in \mathbb{T}$ defined by t .

Example 2.2. For any $\alpha \in \mathbb{R}$, define the *circle rotation by α* to be the map

$$R_\alpha : \mathbb{T} \rightarrow \mathbb{T}, \quad R_\alpha(t) = t + \alpha \pmod{1}.$$

We claim that R_α preserves the Lebesgue measure $m_{\mathbb{T}}$ on the circle. By Theorem A.8, it is enough to prove it for intervals, where it is clear. Alternatively, we may note that Lebesgue measure is a Haar measure on the compact group $\mathbb{T}$, which is invariant under any translation by construction (see Sects. 8.3 and C.2).

Example 2.3. A generalization of Example 2.2 is a rotation on a compact group. Let X be a compact group, and let g be an element of X . Then the map $T_g : X \rightarrow X$ defined by $T_g(x) = gx$ preserves the (left) Haar measure m_X on X . The Haar measure on a locally compact group is described in Appendix C, and may be thought of as the natural generalization of the Lebesgue measure to a general locally compact group.

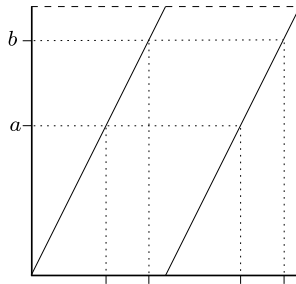


Fig. 2.1 The pre-image of $[a, b]$ under the circle-doubling map

Example 2.4. The *circle-doubling map* is $T_2 : \mathbb{T} \rightarrow \mathbb{T}$, $T_2(t) = 2t \pmod{1}$. We claim that T_2 preserves the Lebesgue measure $m_{\mathbb{T}}$ on the circle. By The-

orem A.8, it is sufficient to check this on intervals, so let $B = [a, b] \subseteq [0, 1]$ be any interval. Then it is easy to check that

$$T_2^{-1}(B) = \left[\frac{a}{2}, \frac{b}{2}\right) \cup \left[\frac{a}{2} + \frac{1}{2}, \frac{b}{2} + \frac{1}{2}\right)$$

is a disjoint union (thinking of a and b as real numbers; see Fig. 2.1), so

$$m_{\mathbb{T}}(T_2^{-1}(B)) = \frac{1}{2}(b - a) + \frac{1}{2}(b - a) = b - a = m_{\mathbb{T}}(B).$$

Notice that the measure-preserving property cannot be seen by studying forward iterates: if I is a small interval, then $T_2(I)$ is an interval* with total length $2(b - a)$.

Example 2.5. Generalizing Example 2.4, let X be a compact abelian group and let $T : X \rightarrow X$ be a surjective endomorphism. Then T preserves the Haar measure m_X on X by the following argument. Define a measure μ on X by $\mu(A) = m_X(T^{-1}A)$. Then, given any $x \in X$ pick y with $T(y) = x$ and notice that

$$\mu(A + x) = m_X(T^{-1}(A + x)) = m_X(T^{-1}A + y) = m_X(T^{-1}A) = \mu(A),$$

so μ is a translation-invariant Borel probability on X (this just means a probability measure defined on the Borel σ -algebra). Since the normalized Haar measure is the unique measure with this property, μ must be m_X , which means that T preserves the Haar measure m_X on X .

One of the ways in which a measure-preserving transformation may be studied is via its induced action on some natural space of functions. Given any function $f : X \rightarrow \mathbb{R}$ and map $T : X \rightarrow X$, write $f \circ T$ for the function defined by $(f \circ T)(x) = f(Tx)$. As usual we write L_{μ}^1 for the space of (equivalence classes of) measurable functions $f : X \rightarrow \mathbb{R}$ with $\int |f| d\mu < \infty$, $\mathcal{L}^{\infty}$ for the space of measurable bounded functions and $\mathcal{L}_{\mu}^1$ for the space of measurable integrable functions (in the usual sense of function, in particular defined everywhere; see Sect. A.3).

Lemma 2.6. *A measure μ on X is T -invariant if and only if*

$$\int f d\mu = \int f \circ T d\mu \tag{2.1}$$

for all $f \in \mathcal{L}^{\infty}$. Moreover, if μ is T -invariant, then (2.1) holds for $f \in L_{\mu}^1$.

PROOF. If (2.1) holds, then for any measurable set B we may take $f = \chi_B$ to see that

* We say that a subset of $\mathbb{T}$ is an interval in $\mathbb{T}$ if it is the image of an interval in $\mathbb{R}$. An interval might therefore be represented in our chosen space of coset representatives $[0, 1]$ by the union of two intervals.

$$\mu(B) = \int \chi_B d\mu = \int \chi_B \circ T d\mu = \int \chi_{T^{-1}B} d\mu = \mu(T^{-1}B),$$

so T preserves μ .

Conversely, if T preserves μ then (2.1) holds for any function of the form χ_B and hence for any simple function (see Sect. A.3). Let f be a non-negative real-valued function in $\mathcal{L}_\mu^1$. Choose a sequence of simple functions (f_n) increasing to f (see Sect. A.3). Then $(f_n \circ T)$ is a sequence of simple functions increasing to $f \circ T$, and so

$$\int f \circ T d\mu = \lim_{n \rightarrow \infty} \int f_n \circ T d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu,$$

showing that (2.1) holds for f . □

One part of ergodic theory is concerned with the structure and classification of measure-preserving transformations. The next definition gives the two basic relationships there may be between measure-preserving transformations⁽¹²⁾.

Definition 2.7. Let $(X, \mathcal{B}_X, \mu, T)$ and $(Y, \mathcal{B}_Y, \nu, S)$ be measure-preserving systems on probability spaces.

- (1) The system $(Y, \mathcal{B}_Y, \nu, S)$ is a *factor* of $(X, \mathcal{B}_X, \mu, T)$ if there are sets X' in $\mathcal{B}_X$ and Y' in $\mathcal{B}_Y$ with $\mu(X') = 1$, $\nu(Y') = 1$, $TX' \subseteq X'$, $SY' \subseteq Y'$ and a measure-preserving map $\phi : X' \rightarrow Y'$ with

$$\phi \circ T(x) = S \circ \phi(x)$$

for all $x \in X'$.

- (2) The system $(Y, \mathcal{B}_Y, \nu, S)$ is *isomorphic* to $(X, \mathcal{B}_X, \mu, T)$ if there are sets X' in $\mathcal{B}_X$, Y' in $\mathcal{B}_Y$ with $\mu(X') = 1$, $\nu(Y') = 1$, $TX' \subseteq X'$, $SY' \subseteq Y'$, and an invertible measure-preserving map $\phi : X' \rightarrow Y'$ with

$$\phi \circ T(x) = S \circ \phi(x)$$

for all $x \in X'$.

In measure theory it is natural to simply ignore null sets, and we will sometimes loosely think of a factor as a measure-preserving map $\phi : X \rightarrow Y$ for which the diagram

$$\begin{array}{ccc} X & \xrightarrow{T} & X \\ \phi \downarrow & & \downarrow \phi \\ Y & \xrightarrow{S} & Y \end{array}$$

is commutative, with the understanding that the map is not required to be defined everywhere.

A factor map

$$(X, \mathcal{B}_X, \mu, T) \longrightarrow (Y, \mathcal{B}_Y, \nu, S)$$

will also be described as an *extension* of $(Y, \mathcal{B}_Y, \nu, S)$. The factor $(Y, \mathcal{B}_Y, \nu, S)$ is called trivial if as a measure space Y comprises a single element; the extension is called trivial if ϕ is an isomorphism of measure spaces.

Example 2.8. Define the $(\frac{1}{2}, \frac{1}{2})$ measure $\mu_{(1/2, 1/2)}$ on the finite set $\{0, 1\}$ by

$$\mu_{(1/2, 1/2)}(\{0\}) = \mu_{(1/2, 1/2)}(\{1\}) = \frac{1}{2}.$$

Let $X = \{0, 1\}^{\mathbb{N}}$ with the infinite product measure $\mu = \prod_{\mathbb{N}} \mu_{(1/2, 1/2)}$ (see Sect. A.2 and Example 2.9 where we will generalize this example). This space is a natural model for the set of possible outcomes of the infinitely repeated toss of a fair coin. The *left shift map* $\sigma : X \rightarrow X$ defined by

$$\sigma(x_0, x_1, \dots) = (x_1, x_2, \dots)$$

preserves μ (since it preserves the measure of the cylinder sets described in Example 2.9). The map $\phi : X \rightarrow \mathbb{T}$ defined by

$$\phi(x_0, x_1, \dots) = \sum_{n=0}^{\infty} \frac{x_n}{2^{n+1}}$$

is measure-preserving from (X, μ) to $(\mathbb{T}, m_{\mathbb{T}})$ and $\phi(\sigma(x)) = T_2(\phi(x))$. The map ϕ has a measurable inverse defined on all but the countable set of dyadic rationals $\mathbb{Z}[\frac{1}{2}]/\mathbb{Z}$, where

$$\mathbb{Z}[\frac{1}{2}] = \{\frac{m}{2^n} \mid m \in \mathbb{Z}, n \in \mathbb{N}\},$$

so this shows that (X, μ, σ) and $(\mathbb{T}, m_{\mathbb{T}}, T_2)$ are measurably isomorphic.

When the underlying space is a compact metric space, the σ -algebra is taken to be the Borel σ -algebra (the smallest σ -algebra containing all the open sets) unless explicitly stated otherwise. Notice that in both Example 2.8 and Example 2.9 the underlying space is indeed a compact metric space (see Sect. A.2).

Example 2.9. The shift map in Example 2.8 is an example of a one-sided Bernoulli shift. A more general⁽¹³⁾ and natural two-sided definition is the following. Consider an infinitely repeated throw of a loaded n -sided die. The possible outcomes of each throw are $\{1, 2, \dots, n\}$, and these appear with probabilities given by the probability vector $\mathbf{p} = (p_1, p_2, \dots, p_n)$ (probability vector means each $p_i \geq 0$ and $\sum_{i=1}^n p_i = 1$), so $\mathbf{p}$ defines a measure $\mu_{\mathbf{p}}$ on the finite sample space $\{1, 2, \dots, n\}$, which is given the discrete topology. The sample space for the die throw repeated infinitely often is

$$\begin{aligned}
X &= \{1, 2, \dots, n\}^{\mathbb{Z}} \\
&= \{x = (\dots, x_{-1}, x_0, x_1, \dots) \mid x_i \in \{1, 2, \dots, n\} \text{ for all } i \in \mathbb{Z}\}.
\end{aligned}$$

The measure on X is the infinite product measure $\mu = \prod_{\mathbb{Z}} \mu_{\mathbf{p}}$, and the σ -algebra $\mathcal{B}$ is the Borel σ -algebra for the compact metric space^{*} X , or equivalently is the product σ -algebra defined below and in Sect. A.2.

A better description of the measure is given via *cylinder sets*. If I is a finite subset of $\mathbb{Z}$, and $\mathbf{a}$ is a map $I \rightarrow \{1, 2, \dots, n\}$, then the cylinder set defined by I and $\mathbf{a}$ is

$$I(\mathbf{a}) = \{x \in X \mid x_j = \mathbf{a}(j) \text{ for all } j \in I\}.$$

It will be useful later to write $x|_I$ for the ordered block of coordinates

$$x_i x_{i+1} \cdots x_{i+s}$$

when $I = \{i, i+1, \dots, i+s\} = [i, i+s]$. The measure μ is uniquely determined by the property that

$$\mu(I(\mathbf{a})) = \prod_{i \in I} p_{\mathbf{a}(i)},$$

and $\mathcal{B}$ is the smallest σ -algebra containing all cylinders (see Sect. A.2 for the details).

Now let σ be the (left) shift on X : $\sigma(x) = y$ where $y_j = x_{j+1}$ for all j in $\mathbb{Z}$. Then σ is μ -preserving and $\mathcal{B}$ -measurable. So $(X, \mathcal{B}, \mu, \sigma)$ is a measure-preserving system, called the *Bernoulli scheme* or *Bernoulli shift* based on $\mathbf{p}$. A measure-preserving system measurably isomorphic to a Bernoulli shift is sometimes called a Bernoulli automorphism.

The next example, which we learned from Doug Lind, gives another example of a measurable isomorphism and reinforces the point that being a probability space is a finiteness property of the measure, rather than a metric boundedness property of the space. The measure μ on $\mathbb{R}$ described in Example 2.10 makes $(\mathbb{R}, \mu)$ into a probability space.

Example 2.10. Consider the 2-to-1 map $T: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$T(x) = \frac{1}{2} \left(x - \frac{1}{x} \right)$$

for $x \neq 0$, and $T(0) = 0$. For any L^1 function f , the substitution $y = T(x)$ shows that

* The topology on X is simply the product topology, which is also the metric topology given by the metric defined by $\mathbf{d}(x, y) = 2^{-k}$ where

$$k = \max\{j \mid x_i = y_i \text{ for } |j| \leq k\}$$

if $x \neq y$ and $\mathbf{d}(x, x) = 0$. In this metric, points are close together if they agree on a large block of indices around $0 \in \mathbb{Z}$.

$$\int_{-\infty}^{\infty} f(T(x)) \frac{dx}{\pi(1+x^2)} = \int_{-\infty}^{\infty} f(y) \frac{dy}{\pi(1+y^2)}$$

(in this calculation, note that T is only injective when restricted to $(0, \infty)$ or $(-\infty, 0)$). It follows by Lemma 2.6 that T preserves the probability measure μ defined by

$$\mu([a, b]) = \int_a^b \frac{dx}{\pi(1+x^2)}.$$

The map $\phi(x) = \frac{1}{\pi} \arctan(x) + \frac{1}{2}$ from $\mathbb{R}$ to $\mathbb{T}$ is an invertible measure-preserving map from $(\mathbb{R}, \mu)$ to $(\mathbb{T}, m_{\mathbb{T}})$ where $m_{\mathbb{T}}$ denotes the Lebesgue measure on $\mathbb{T}$ (notice that the image of ϕ is the subset $(0, 1) \subseteq \mathbb{T}$, but this is an invertible map in the measure-theoretic sense).

Define the map $T_2 : \mathbb{T} \rightarrow \mathbb{T}$ by $T_2(x) = 2x \pmod{1}$ as in Example 2.4. The map ϕ is a measurable isomorphism from $(\mathbb{R}, \mu, T)$ to $(\mathbb{T}, m_{\mathbb{T}}, T_2)$. Example 2.8 shows in turn that $(\mathbb{R}, \mu, T)$ is isomorphic to the one-sided full 2-shift.

It is often more convenient to work with an invertible measure-preserving transformation as in Example 2.9 instead of a non-invertible transformation as in Examples 2.4 and 2.8. Exercise 2.1.7 gives a general construction of an invertible system from a non-invertible one.

Exercises for Sect. 2.1

Exercise 2.1.1. Show that the space $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}})$ is isomorphic as a measure space to $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2})$.

Exercise 2.1.2. Show that the measure-preserving system $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}}, T_4)$, where $T_4(x) = 4x \pmod{1}$, is measurably isomorphic to the product system $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2}, T_2 \times T_2)$.

Exercise 2.1.3. For a map $T : X \rightarrow X$ and sets $A, B \subseteq X$, prove the following.

- $\chi_A(T(x)) = \chi_{T^{-1}(A)}(x)$;
- $T^{-1}(A \cap B) = T^{-1}(A) \cap T^{-1}(B)$;
- $T^{-1}(A \cup B) = T^{-1}(A) \cup T^{-1}(B)$;
- $T^{-1}(A \triangle B) = T^{-1}(A) \triangle T^{-1}(B)$.

Which of these properties also hold with the pre-image under T^{-1} replaced by the forward image under T ?

Exercise 2.1.4. What happens to Example 2.5 if the map $T : X \rightarrow X$ is only required to be a continuous homomorphism?

Exercise 2.1.5. (a) Find a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with a non-trivial factor map $\phi : X \rightarrow X$.
 (b) Find an invertible measure-preserving system $(X, \mathcal{B}, \mu, T)$ with a non-trivial factor map $\phi : X \rightarrow X$.

Exercise 2.1.6. Prove that the circle rotation R_α from Example 2.2 is not measurably isomorphic to the circle-doubling map T_2 from Example 2.4.

Exercise 2.1.7. Let $X = (X, \mathcal{B}, \mu, T)$ be any measure-preserving system. A sub- σ -algebra $\mathcal{A} \subseteq \mathcal{B}_X$ with $T^{-1}\mathcal{A} = \mathcal{A}$ modulo μ is called a *T-invariant sub- σ -algebra*. Show that the system $\tilde{X} = (\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu}, \tilde{T})$ defined by

- $\tilde{X} = \{x \in X^{\mathbb{Z}} \mid x_{k+1} = T(x_k) \text{ for all } k \in \mathbb{Z}\};$
- $(\tilde{T}(x))_k = x_{k+1}$ for all $k \in \mathbb{Z}$ and $x \in \tilde{X}$;
- $\tilde{\mu}(\{x \in \tilde{X} \mid x_0 \in A\}) = \mu(A)$ for any $A \in \mathcal{B}$, and $\tilde{\mu}$ is invariant under $\tilde{T}$;
- $\tilde{\mathcal{B}}$ is the smallest $\tilde{T}$ -invariant σ -algebra for which the map $\pi : x \mapsto x_0$ from $\tilde{X}$ to X is measurable;

is an invertible measure-preserving system, and that the map $\pi : x \mapsto x_0$ is a factor map. The system $\tilde{X}$ is called the *invertible extension* of X .

Exercise 2.1.8. Show that the invertible extension $\tilde{X}$ of a measure-preserving system X constructed in Exercise 2.1.7 has the following universal property. For any extension

$$\phi : (Y, \mathcal{B}_Y, \nu, S) \rightarrow (X, \mathcal{B}_X, \mu, T)$$

for which S is invertible, there exists a unique map

$$\tilde{\phi} : (Y, \mathcal{B}_Y, \nu, S) \rightarrow (\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu}, \tilde{T})$$

for which $\phi = \pi \circ \tilde{\phi}$.

Exercise 2.1.9. (a) Show that the invertible extension of the circle-doubling map from Example 2.4,

$$X_2 = \{x \in \mathbb{T}^{\mathbb{Z}} \mid x_{k+1} = T_2 x_k \text{ for all } k \in \mathbb{Z}\},$$

is a compact abelian group with respect to the coordinate-wise addition defined by $(x + y)_k = x_k + y_k$ for all $k \in \mathbb{Z}$, and the topology inherited from the product topology on $\mathbb{T}^{\mathbb{Z}}$.

(b) Show that the diagonal embedding $\delta(r) = (r, r)$ embeds $\mathbb{Z}[\frac{1}{2}]$ as a discrete subgroup of $\mathbb{R} \times \mathbb{Q}_2$, and that $X_2 \cong \mathbb{R} \times \mathbb{Q}_2 / \delta(\mathbb{Z}[\frac{1}{2}]) \cong \mathbb{R} \times \mathbb{Z}_2 / \delta(\mathbb{Z})$ as compact abelian groups (see Appendix C for the definition of $\mathbb{Q}_p$ and $\mathbb{Z}_p$). In particular, the map $\tilde{T}_2$ (which may be thought of as the left shift on X_2 , or as the map that doubles in each coordinate) is conjugate to the map

$$(s, r) + \delta(\mathbb{Z}[\frac{1}{2}]) \mapsto (2s, 2r) + \delta(\mathbb{Z}[\frac{1}{2}])$$